

Paper 1 - Set 11 - Solutions

Generated practice paper: answer key and worked solutions.

Answer key

Question	Answer	Topic
1	C	Indices and surds
2	D	Simultaneous equations and inequalities
3	B	Circles and circle theorems
4	D	Integration
5	F	Ratio, proportion and percentages
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10	E	Indices and surds
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19	F	Integration
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Worked solutions

Question 1 IND-15 · Indices and surds · spec MM1.1

Given that $2^x = 5$ and $2^y = 20$, find the value of 2^{2x-y} .

Answer: C

$$2^{2x-y} = \frac{(2^x)^2}{2^y} = \frac{25}{20} = \frac{5}{4}.$$

Full method - how and why

You are given two exponentials and asked for a third. Express the target in terms of the given ones using index laws - you never need to find x or y .

1 Split the index: $2^{2x-y} = \frac{2^{2x}}{2^y}$.

why: Subtraction in the index is division - this is what lets you use the two given facts separately.

2 Write $2^{2x} = (2^x)^2 = 5^2 = 25$.

why: A coefficient in the index is a power of the whole exponential: $2^{2x} = (2^x)^2$, not 2×2^x .

3 Substitute: $\frac{25}{20} = \frac{5}{4}$.

why: $2^y = 20$ is given directly.

Where it goes wrong

- Using $2^{2x} = 2 \times 2^x = 10$, giving $\frac{1}{2}$.
- Trying to solve for $x = \log_2 5$ and $y = \log_2 20$ first - correct but slow, and it invites arithmetic slips.

Takeaway Rewrite the target index as sums and multiples of the given ones; substitution then beats solving for the unknowns.

Question 2 SIM-07 · Simultaneous equations and inequalities · spec MM1.5

The inequality $x^2 + bx + 9 > 0$ holds for all real x . Find the complete set of values of b .

Answer: D

Need discriminant negative: $b^2 - 36 < 0$, so $-6 < b < 6$.

Full method - how and why

'True for all x ' for an upward parabola means it never touches the axis: discriminant strictly negative.

1 Note the leading coefficient is $1 > 0$, so the parabola opens upwards.

why: For > 0 everywhere you need both the right shape and no real roots; here the shape is automatic.

2 Require no real roots: $b^2 - 4(1)(9) < 0$.

why: If there were real roots the curve would touch or cross the axis and the expression would fail to be positive there.

3 Solve: $b^2 < 36 \Rightarrow -6 < b < 6$.

why: $b^2 < 36$ means $|b| < 6$, a single interval.

Where it goes wrong

- Solving $b^2 < 36$ as two outer intervals - that is the solution of $b^2 > 36$.
- Using $\leq$: at $b = \pm 6$ the curve touches the axis, where the expression equals 0 and is not > 0 .

Takeaway 'Positive for all x ' $\Leftrightarrow a > 0$ and $b^2 - 4ac < 0$; $x^2 < k$ gives one interval, $x^2 > k$ gives two.

Question 3 CIR-13 · Circles and circle theorems · spec MM3.2

The circle $x^2 + y^2 = r^2$ and the line $3x + 4y = 20$ have exactly one point in common. Find r .

Answer: B

Distance from origin to the line $= \frac{20}{\sqrt{9+16}} = 4$. Tangency means $r = 4$.

Full method - how and why

One common point means tangency, so the radius equals the distance from the centre to the line.

- 1 The circle is centred at the origin, so compute the distance from $(0, 0)$ to $3x + 4y = 20$.

why: Centre-at-origin makes the distance formula especially clean.

- 2 Distance $= \frac{|3(0) + 4(0) - 20|}{\sqrt{3^2 + 4^2}} = \frac{20}{5} = 4$.

why: $\sqrt{9+16} = 5$ the familiar 3-4-5 triple.

- 3 Tangency requires r equal to this distance: $r = 4$.

why: If r were larger the line would cut the circle twice; smaller, and it would miss.

Where it goes wrong

- Dividing by $\sqrt{20}$ or by 7 instead of 5.
- Answering 20 or $\sqrt{20}$ by stopping too early.

Takeaway Distance from the origin to $ax + by = c$ is $\frac{|c|}{\sqrt{a^2 + b^2}}$; tangency sets it equal to r .

Question 4 INT-18 · Integration · spec MM7.3

Let $I = \int_0^1 (x^2 + 1) dx$ and $J = \int_0^1 (x + 1)^2 dx$. Which of the following is true?

Answer: D

$$I = \frac{1}{3} + 1 = \frac{4}{3}, J = \int_0^1 (x^2 + 2x + 1) dx = \frac{1}{3} + 1 + 1 = \frac{7}{3}, J - I = 1.$$

Full method - how and why

Two similar-looking integrals. Expand the second - $(x + 1)^2 \neq x^2 + 1$ - and compare.

- 1 Compute I : $\int_0^1 (x^2 + 1) dx = \left[\frac{x^3}{3} + x \right]_0^1 = \frac{1}{3} + 1 = \frac{4}{3}$.

why: Term by term.

- 2 Expand the second integrand: $(x + 1)^2 = x^2 + 2x + 1$.

why: The cross term $2x$ is the whole difference between the two integrals.

- 3 Compute J : $\left[\frac{x^3}{3} + x^2 + x \right]_0^1 = \frac{1}{3} + 1 + 1 = \frac{7}{3}$.

why: The extra $\int_0^1 2x dx = 1$.

- 4 Compare: $J - I = \frac{7}{3} - \frac{4}{3} = 1$, and $J > I$.

why: Equivalently, the difference is exactly $\int_0^1 2x dx = 1$.

Where it goes wrong

- Assuming $I = J$ because the integrands 'look the same'.
- Arithmetic giving $J - I = \frac{4}{3}$ - an offered option.

Takeaway $(x + 1)^2 = x^2 + 2x + 1$; the difference between the two integrals is the integral of the cross term.

Question 5 RAT-05 · Ratio, proportion and percentages · spec M3.11

A bacteria population doubles every 3 hours. By what factor does it grow in 24 hours?

Answer: F

$$24/3 = 8 \text{ doublings: } 2^8 = 256.$$

Full method - how and why

Repeated doubling. Count the doublings, then raise 2 to that power.

- 1 Count the doubling periods: $24 \div 3 = 8$.
why: Each 3-hour block doubles the population.
- 2 Each doubling multiplies by 2, so eight of them multiply by 2^8 .
why: Repeated multiplication is exponentiation - not multiplication by 8.
- 3 $2^8 = 256$.
why: Worth knowing the powers of 2 up to $2^{10} = 1024$.

Where it goes wrong

- Multiplying by 8 or computing $2 \times 8 = 16$ - both offered.
- Using $2^6 = 64$ by dividing 24 by 4.

Takeaway Growth factor = (factor per period)^{number of periods}.

Question 6 DIF-12 · Differentiation · spec MM6.3

The line $y = 4x + c$ is a tangent to $y = x^2 + 2x + 5$. Find c .

Answer: A

$$2x + 2 = 4 \Rightarrow x = 1, y = 8. 8 = 4 + c \Rightarrow c = 4.$$

Full method - how and why

Tangency via calculus: the gradients must match, which locates the point of contact.

- 1 Differentiate the curve: $y' = 2x + 2$.
why: The tangent's gradient equals the curve's gradient at the contact point.
- 2 Set it equal to the line's gradient: $2x + 2 = 4$, so $x = 1$.
why: This finds **where** the tangent touches.
- 3 Find the y-value on the curve: $y(1) = 1 + 2 + 5 = 8$.
why: The contact point is on the curve.
- 4 The point must also be on the line: $8 = 4(1) + c$, so $c = 4$.
why: Substituting the contact point into the line's equation gives c .

Where it goes wrong

- Using the discriminant route but mis-expanding and getting $c = 1$ or 5 (both offered).
- Substituting into the line the y-value from the wrong x .

Takeaway Tangency: equal gradients give the x ; equal y-values give the constant.

Question 7 NUM-20 · Number, units and estimation · spec M2.11

Find the exact value of $(\sqrt{5} - 1)^2 + (\sqrt{5} + 1)^2$.

Answer: A

$$(6 - 2\sqrt{5}) + (6 + 2\sqrt{5}) = 12.$$

Full method - how and why

A sum of conjugate squares. The surd terms cancel, so the answer is rational.

1 Expand the first: $(\sqrt{5} - 1)^2 = 5 - 2\sqrt{5} + 1 = 6 - 2\sqrt{5}$.
why: $(a - b)^2 = a^2 - 2ab + b^2$ with $(\sqrt{5})^2 = 5$.

2 Expand the second: $(\sqrt{5} + 1)^2 = 6 + 2\sqrt{5}$.
why: Only the middle sign differs.

3 Add: the $\mp 2\sqrt{5}$ terms cancel, leaving 12.
why: Conjugates always cancel their surd parts when added.

Where it goes wrong

- Dropping the cross terms: here it happens to give 12 anyway, but the same slip fails whenever the two brackets are not conjugates.
- Adding the surd terms instead of cancelling, giving $12 + 4\sqrt{5}$ - an offered option.

Takeaway $(a + b)^2 + (a - b)^2 = 2(a^2 + b^2)$: here $2(5 + 1) = 12$, with no surds surviving.

Question 8 SEQ-18 · Sequences and series · spec MM2.1

A sequence is defined by $u_1 = 1$ and $u_{n+1} = u_n + 2n + 1$. Which of the following is an expression for u_n ?

Answer: F

$$u_2 = 4, u_3 = 9, u_4 = 16; u_n = n^2. \text{ Check: } (n + 1)^2 = n^2 + 2n + 1.$$

Full method - how and why

A recurrence that generates a familiar sequence. Compute terms, recognise the pattern, then verify it satisfies the rule.

1 Generate terms: $u_1 = 1, u_2 = 1 + 3 = 4, u_3 = 4 + 5 = 9, u_4 = 9 + 7 = 16$.
why: The increments 3, 5, 7 are the odd numbers - a strong hint.

2 Recognise 1, 4, 9, 16 as the squares: conjecture $u_n = n^2$.
why: Pattern-spotting is legitimate here because the options let you verify.

3 Verify against the recurrence: if $u_n = n^2$ then $u_n + 2n + 1 = n^2 + 2n + 1 = (n + 1)^2 = u_{n+1}$ ✓.
why: This is the crucial step: checking the formula satisfies the rule turns a guess into a proof.

Where it goes wrong

- Choosing $2n - 1$ (the increments) instead of the terms themselves.
- Testing only u_1 , which several options match.

Takeaway Spot the pattern from a few terms, then prove it by substituting into the recurrence.

Question 9 QUA-23 · Quadratic functions · spec MM1.3

The line $y = mx$ meets the curve $y = x^2 + 9$ at two distinct points. Find the complete set of values of m .

Answer: B

$$x^2 - mx + 9 = 0 \text{ needs } m^2 - 36 > 0, \text{ so } m < -6 \text{ or } m > 6.$$

Full method - how and why

A line meeting a curve twice: substitute and demand a positive discriminant.

- 1 Substitute $y = mx$ into $y = x^2 + 9$: $mx = x^2 + 9$.

why: Intersections are the common solutions, found by equating the two expressions for y .

- 2 Rearrange: $x^2 - mx + 9 = 0$.

why: Standard form is needed before the discriminant can be read off.

- 3 Two distinct intersections $\Rightarrow$ discriminant > 0 : $m^2 - 36 > 0$.

why: Here $b = -m$, and $b^2 = m^2$ regardless of sign.

- 4 Solve $m^2 > 36$: $m < -6$ or $m > 6$.

why: Two intervals, since $m^2 > 36$ means $|m| > 6$; geometrically, steep lines in either direction cut the parabola twice.

Where it goes wrong

- Answering $-6 < m < 6$ (the region where the line misses the parabola entirely).
- Writing the discriminant as $-m^2 - 36$ by mishandling the sign of b .

Takeaway b^2 kills the sign of b : conditions like this are almost always symmetric in $\pm m$.

Question 10 IND-05 · Indices and surds · spec MM1.2

Simplify $\sqrt{75} - \sqrt{27} + \sqrt{12}$.

Answer: E

$$\sqrt{75} = 5\sqrt{3}, \sqrt{27} = 3\sqrt{3}, \sqrt{12} = 2\sqrt{3}, \text{ so the total is } (5 - 3 + 2)\sqrt{3} = 4\sqrt{3}.$$

Full method - how and why

Simplifying a sum of surds. They can only be combined once each is written as a multiple of the **same** surd.

- 1 Pull out the largest square factor from each: $\sqrt{75} = \sqrt{25 \times 3} = 5\sqrt{3}$, $\sqrt{27} = \sqrt{9 \times 3} = 3\sqrt{3}$,
 $\sqrt{12} = \sqrt{4 \times 3} = 2\sqrt{3}$.

why: $\sqrt{ab} = \sqrt{a}\sqrt{b}$, so a perfect-square factor comes out as a whole number. All three reduce to multiples of $\sqrt{3}$, which is why the sum simplifies at all.

- 2 Collect like terms: $5\sqrt{3} - 3\sqrt{3} + 2\sqrt{3} = 4\sqrt{3}$.

why: Once every term has the same surd part, $\sqrt{3}$ behaves exactly like an algebraic x .

Where it goes wrong

- Adding under one root sign: $\sqrt{75} - \sqrt{27} \neq \sqrt{48}$. Roots do not distribute over $+$ or $-$.
- Using $\sqrt{75} = 3\sqrt{5}$ (wrong square factor) - always take out the **largest** square.

Takeaway Surds add only when their irrational parts match, so simplify each one fully before trying to combine.

Question 11 CIR-12 · Circles and circle theorems · spec MM3.2

The circle C has equation $x^2 + y^2 - 4x - 6y + 9 = 0$. Which of the following lines does not meet C ?

Answer: D

Centre $(2, 3)$, radius 2, so the circle spans $0 \leq x \leq 4$ and $1 \leq y \leq 5$. The line $y = 0$ is 3 from the centre, more than the radius, so it misses; $x = 0$, $x = 4$ and $y = 5$ touch and $x = 2$ passes through the centre.

Full method - how and why

Which line misses the circle: compare each line's distance from the centre with the radius. Horizontal and vertical lines make this trivial.

- 1 Complete the square: $(x - 2)^2 + (y - 3)^2 = -9 + 4 + 9 = 4$, so centre $(2, 3)$ and radius 2.

why: Always convert to centre-radius form first.

- 2 The circle therefore spans x from 0 to 4 and y from 1 to 5.

why: Centre $\pm$ radius in each direction gives the bounding box, whose edges are tangent lines.

- 3 Check each line: $x = 0$, $x = 4$ and $y = 5$ are tangents (distance exactly 2); $x = 2$ passes through the centre.

why: For a vertical line $x = k$, the distance from the centre is $|k - 2|$.

- 4 $y = 0$ is at distance $|0 - 3| = 3 > 2$, so it misses the circle.

why: Distance greater than the radius means no intersection.

Where it goes wrong

- Assuming a tangent line 'does not meet' the circle - touching counts as meeting.
- Getting $r^2 = 9$ by mishandling the $+9$ constant.

Takeaway For horizontal/vertical lines just compare the centre coordinate with the radius; tangency still counts as meeting.

Question 12 SEQ-13 · Sequences and series · spec MM2.2

An arithmetic sequence has $u_3 = 10$ and $u_8 = 25$. Find the smallest value of n for which $u_n > 100$.

Answer: C

$$5d = 15 \Rightarrow d = 3, a = 4. u_n = 4 + 3(n - 1) = 3n + 1 > 100 \Rightarrow n > 33, \text{ so } n = 34.$$

Full method - how and why

Two terms give a and d ; then solve an inequality for n and round to the correct integer.

- 1 Subtract the two facts: $u_8 - u_3 = 5d = 25 - 10 = 15$, so $d = 3$.

why: From u_3 to u_8 is five steps, so the difference is $5d$ - count the gaps, not the terms.

- 2 Find a : $u_3 = a + 2d = 10$, so $a = 10 - 6 = 4$.

why: The third term is $a + 2d$, not $a + 3d$.

- 3 Write the general term: $u_n = 4 + 3(n - 1) = 3n + 1$.

why: Simplify before solving - it avoids bracket errors.

- 4 Solve $3n + 1 > 100 \Rightarrow n > 33$, so the smallest integer is $n = 34$.

why: Strict inequality: $n = 33$ gives exactly 100, which is not greater than 100.

Where it goes wrong

- Answering 33, where $u_{33} = 100$ exactly.
- Using $u_8 - u_3 = 8d - 3d$ correctly but then $u_3 = a + 3d$ incorrectly.

Takeaway u_p to u_q spans $q - p$ common differences; and 'greater than' excludes the boundary case.

Question 13 QUA-08 · Quadratic functions · spec MM1.3

The equation $kx^2 + 4x + k = 0$ has no real roots. Find the complete set of values of k .

Answer: C

Need $16 - 4k^2 < 0$, i.e. $k^2 > 4$, so $k < -2$ or $k > 2$. (If $k = 0$ the equation is linear with one root, so $k = 0$ is excluded anyway.)

Full method - how and why

'No real roots' is the discriminant < 0 . The extra subtlety is the coefficient k appearing in the x^2 term, so $k = 0$ needs a thought.

- 1 Write the discriminant: $4^2 - 4(k)(k) = 16 - 4k^2$.

why: Here $a = k$ and $c = k$ - both the first and last coefficients are the parameter.

- 2 Set it negative: $16 - 4k^2 < 0 \Rightarrow k^2 > 4$.

why: No real roots $\Leftrightarrow b^2 - 4ac < 0$.

- 3 Solve $k^2 > 4$: $k < -2$ or $k > 2$.

why: $k^2 > 4$ means $|k| > 2$, which is two separate intervals, not $-2 < k < 2$.

- 4 Check $k = 0$ separately: the equation becomes $4x = 0$, which has the root $x = 0$.

why: If $k = 0$ it is not a quadratic at all, so the discriminant test does not apply - but $k = 0$ is excluded by the answer anyway.

Where it goes wrong

- Solving $k^2 > 4$ as $-2 < k < 2$ - that is the solution of $k^2 < 4$.
- Ignoring the possibility $k = 0$ entirely (harmless here, fatal in questions where 0 lies inside the answer set).

Takeaway $k^2 > c$ gives two outer intervals; and whenever the leading coefficient contains the parameter, check the value that makes it zero.

Question 14 GEO-05 · Angles, polygons, congruence and vectors · spec M5.5

Two similar triangles have corresponding sides of length 6 cm and 9 cm. The smaller triangle has area 20 cm². Find the area of the larger triangle.

Answer: D

Area scale factor $\left(\frac{9}{6}\right)^2 = \frac{9}{4}$: $20 \times \frac{9}{4} = 45$.

Full method - how and why

Similar shapes: areas scale as the **square** of the length ratio.

- 1 Length scale factor: $\frac{9}{6} = \frac{3}{2}$.

why: From the smaller to the larger, so the factor is greater than 1.

- 2 Area scale factor: $\left(\frac{3}{2}\right)^2 = \frac{9}{4}$.

why: Both dimensions scale, so the area scales by the square.

- 3 Multiply: $20 \times \frac{9}{4} = 45$ cm².

why: $20 \div 4 = 5$, then $\times 9$.

Where it goes wrong

- Using the length factor directly: $20 \times \frac{3}{2} = 30$ - an offered option.
- Cubing (that would be for volumes), giving 67.5.

Takeaway Area ratio = (length ratio)²; check the direction of the scale factor.

Question 15 BIN-08 · Binomial expansion · spec MM2.4

The first three terms in the expansion of $(1 + ax)^n$ in ascending powers of x are $1 + 12x + 54x^2$. Find n .

Answer: D

$na = 12$ and $\frac{n(n-1)}{2}a^2 = 54$. Dividing the second by the square of the first: $\frac{n-1}{2n} = \frac{54}{144} = \frac{3}{8}$, so $8n - 8 = 6n$ and $n = 4$ (then $a = 3$; check $\binom{4}{2} \cdot 9 = 54$).

Full method - how and why

Two coefficients, two unknowns (a and n). Dividing the equations cleverly eliminates a .

- 1 Write the first two coefficients: $\binom{n}{1}a = na = 12$ and $\binom{n}{2}a^2 = \frac{n(n-1)}{2}a^2 = 54$.

why: The expansion of $(1 + ax)^n$ starts $1 + nax + \frac{n(n-1)}{2}a^2x^2$.

- 2 Square the first equation: $n^2a^2 = 144$.

why: Squaring makes the power of a match the second equation, so dividing will cancel it.

- 3 Divide the second by this: $\frac{n(n-1)a^2/2}{n^2a^2} = \frac{n-1}{2n} = \frac{54}{144} = \frac{3}{8}$.

why: Both a^2 and one factor of n cancel.

- 4 Cross-multiply: $8(n-1) = 6n \Rightarrow 8n - 8 = 6n \Rightarrow n = 4$.

why: Then $a = 3$; check $\binom{4}{2} \times 9 = 6 \times 9 = 54 \checkmark$.

Where it goes wrong

- Dividing the second equation by the first (not its square), which leaves an a behind and stalls.
- Using $\binom{n}{2} = \frac{n^2}{2}$.

Takeaway To eliminate a constant that appears to different powers, raise one equation to the matching power first, then divide.

Question 16 POL-10 · Polynomials, factor and remainder theorems · spec MM1.6

Given that $x^3 + px + q$ is divisible by $(x - 2)^2$, find the value of q .

Answer: A

$x^3 + px + q = (x - 2)^2(x - r)$. Comparing x^2 coefficients: $-4 - r = 0 \Rightarrow r = -4$. Constant term: $(4)(4) = 16 = q$. (Also $p = 4 + 4r = -12$.)

Full method - how and why

A repeated factor $(x - 2)^2$ means $x = 2$ is a double root. Comparing coefficients in the factorised form gives q without any calculus.

- 1 Write $x^3 + px + q = (x - 2)^2(x - r)$.

why: A cubic with a repeated root 2 and one other root r must take this form (leading coefficient 1).

- 2 Compare x^2 coefficients. The left-hand side has none, so the coefficient is 0; expanding the right gives $-(4 + r)$... more precisely $(x^2 - 4x + 4)(x - r)$ has x^2 coefficient $-r - 4$.

why: The **absence** of a term is information: its coefficient is zero.

- 3 So $-r - 4 = 0$, giving $r = -4$.

why: One comparison is enough to find the third root.

- 4 Constant term: $4 \times (-r) = 4 \times 4 = 16$, so $q = 16$.

why: The constant of $(x - 2)^2(x - r)$ is $4 \times (-r)$; with $r = -4$ that is 16.

Where it goes wrong

- Forgetting that the missing x^2 term forces a condition, and trying to solve with only the constant term.
- Sign slip on $-r$, giving $q = -16$.

Takeaway A missing power in a polynomial is a coefficient equal to zero - often the most useful equation available.

Question 17 TRI-15 · Triangles, radians and sectors · spec MM4.1

Triangle ABC has $AB = 7$, $BC = 8$, $CA = 9$. Find the area of the triangle.

Answer: D

$$\cos A = \frac{49 + 81 - 64}{2 \cdot 7 \cdot 9} = \frac{66}{126} = \frac{11}{21}, \text{ so } \sin A = \frac{\sqrt{441 - 121}}{21} = \frac{\sqrt{320}}{21} = \frac{8\sqrt{5}}{21}. \text{ Area} = \frac{1}{2} \cdot 7 \cdot 9 \cdot \frac{8\sqrt{5}}{21} = 12\sqrt{5}.$$

Full method - how and why

Three sides, area wanted. Find a cosine, convert to a sine, then use $\frac{1}{2}ab\sin C$ - all without a calculator.

1 Find $\cos A$ where A is between the sides 7 and 9: $\cos A = \frac{49 + 81 - 64}{2(7)(9)} = \frac{66}{126} = \frac{11}{21}$.

why: The side opposite A is $BC = 8$, so that is the one subtracted.

2 Convert to $\sin A$ using $\sin^2 + \cos^2 = 1$: $\sin A = \sqrt{1 - \frac{121}{441}} = \frac{\sqrt{320}}{21}$.

why: Keep the denominator 21 throughout: $1 = \frac{441}{441}$.

3 Simplify the surd: $\sqrt{320} = \sqrt{64 \times 5} = 8\sqrt{5}$, so $\sin A = \frac{8\sqrt{5}}{21}$.

why: Pull out the largest square factor.

4 Area $= \frac{1}{2}(7)(9)\sin A = \frac{63}{2} \times \frac{8\sqrt{5}}{21} = 12\sqrt{5}$.

why: $\frac{63 \times 8}{2 \times 21} = \frac{504}{42} = 12$.

Where it goes wrong

- Taking $\sin A$ positive is right (angles in a triangle are between 0° and 180°), but forgetting to square $\cos A$ before subtracting.
- Arithmetic slips in $\sqrt{320}$, e.g. $16\sqrt{5}$, doubling the area.

Takeaway Three sides $\rightarrow$ cos from the cosine rule $\rightarrow$ sin from the Pythagorean identity $\rightarrow$ area from $\frac{1}{2}ab\sin C$.

Question 18 TRG-14 · Trigonometric functions and equations · spec MM4.6

Solve $12\cos^2 x + 6\sin x - 10 = 2$ for $0^\circ < x < 360^\circ$. Which of the following is the complete set of solutions?

Answer: F

$$12(1 - \sin^2 x) + 6\sin x - 12 = 0 \Rightarrow 12\sin^2 x - 6\sin x = 0 \Rightarrow 6\sin x(2\sin x - 1) = 0. \sin x = 0 \text{ gives } 180^\circ \text{ (the endpoints } 0^\circ \text{ and } 360^\circ \text{ are excluded); } \sin x = \frac{1}{2} \text{ gives } 30^\circ \text{ and } 150^\circ.$$

Full method - how and why

Another quadratic in $\sin x$, but this one has no constant term, so factorising gives $\sin x = 0$ as well - and the interval is open.

1 Replace $\cos^2 x$: $12(1 - \sin^2 x) + 6\sin x - 10 = 2$.

why: Convert to a single trig function.

2 Tidy: $12 - 12\sin^2 x + 6\sin x - 12 = 0 \Rightarrow -12\sin^2 x + 6\sin x = 0 \Rightarrow 6\sin x(2\sin x - 1) = 0$.

why: The constants cancel completely, leaving a **factorisable** expression with no constant term - do not divide by $\sin x$, or you lose solutions.

3 $\sin x = 0$ gives $x = 180^\circ$ within the open interval (0° and 360° are excluded).

why: The interval is $0^\circ < x < 360^\circ$, strict at both ends - so the endpoints are out but 180° is in.

4 $\sin x = \frac{1}{2}$ gives $x = 30^\circ$ and 150° .

why: First and second quadrants.

5 Complete set: $30^\circ, 150^\circ, 180^\circ$.

why: Three solutions - the 180° is what distinguishes the correct option.

Where it goes wrong

- Dividing by $\sin x$ and losing 180° , giving just $30^\circ, 150^\circ$ - an offered option.
- Including 0° and 360° , which the strict inequalities exclude.

Takeaway Never divide by a trig function: factorise, so the zero solutions survive; then read the interval's endpoints carefully.

Question 19 INT-01 · Integration · spec MM7.2

Evaluate $\int_1^4 \frac{3x^2 - 5}{\sqrt{x}} dx$.

Answer: F

$$\begin{aligned} \text{Integrand} &= 3x^{3/2} - 5x^{-1/2}; \text{ integral} \\ &= \left[\frac{6}{5}x^{5/2} - 10x^{1/2} \right]_1^4 = \left(\frac{6 \cdot 32}{5} - 20 \right) - \left(\frac{6}{5} - 10 \right) = \frac{192}{5} - 20 - \frac{6}{5} + 10 = \frac{186}{5} - 10 = \frac{136}{5}. \end{aligned}$$

Full method - how and why

You can only integrate powers of x , so split the fraction first. Then the limits are straightforward - if the fractions are handled carefully.

1 Split the fraction: $\frac{3x^2 - 5}{\sqrt{x}} = 3x^2x^{-1/2} - 5x^{-1/2} = 3x^{3/2} - 5x^{-1/2}$.
why: Dividing by $\sqrt{x}$ subtracts $\frac{1}{2}$ from each index.

2 Integrate: $\int 3x^{3/2} dx = \frac{3x^{5/2}}{5/2} = \frac{6}{5}x^{5/2}$ and $\int 5x^{-1/2} dx = 10x^{1/2}$.
why: Add one to the index and divide by the new index: $\frac{3}{2} + 1 = \frac{5}{2}$, and $-\frac{1}{2} + 1 = \frac{1}{2}$ so dividing by $\frac{1}{2}$ doubles.

3 Evaluate at $x = 4$: $4^{5/2} = 32$ and $4^{1/2} = 2$, giving $\frac{192}{5} - 20$.
why: $4^{5/2} = (\sqrt{4})^5 = 2^5$.

4 Evaluate at $x = 1$: $\frac{6}{5} - 10$. Subtract: $\left(\frac{192}{5} - 20\right) - \left(\frac{6}{5} - 10\right) = \frac{186}{5} - 10 = \frac{136}{5}$.
why: $\frac{192-6}{5} = \frac{186}{5}$, then $-20 + 10 = -10 = -\frac{50}{5}$.

Where it goes wrong

- Forgetting the lower limit entirely, giving $\frac{186}{5}$ - an offered option.
- Dividing by $\frac{5}{2}$ incorrectly: $\frac{3}{5/2} = \frac{6}{5}$, not $\frac{15}{2}$.

Takeaway Split fractions into powers of x before integrating; $4^{5/2} = 32$ type evaluations are worth doing slowly.

Question 20 NUM-09 · Number, units and estimation · spec M2.3

How many positive divisors does 360 have?

Answer: A

$$360 = 2^3 \cdot 3^2 \cdot 5, \text{ so the number of divisors is } (3+1)(2+1)(1+1) = 24.$$

Full method - how and why

Counting divisors from the prime factorisation: each prime's exponent can be chosen independently.

1 Factorise: $360 = 8 \times 45 = 2^3 \times 3^2 \times 5$.

why: Split into convenient factors and factorise each.

2 A divisor has the form $2^a 3^b 5^c$ with $0 \leq a \leq 3$, $0 \leq b \leq 2$, $0 \leq c \leq 1$.

why: Any such combination divides 360, and every divisor has this form uniquely.

3 Count the choices: $4 \times 3 \times 2 = 24$.

why: Each exponent has (power + 1) possible values, because 0 is allowed.

Where it goes wrong

- Multiplying the exponents ($3 \times 2 \times 1 = 6$) instead of (exponent + 1).
- Forgetting that 1 and 360 are themselves divisors - the '+1's account for them.

Takeaway Number of divisors of $p^a q^b r^c$ is $(a+1)(b+1)(c+1)$.